Original Write Histories and a Coupled Source–Field Action

Original source spectrum, six received clocks, and history-preserving computation

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Abstract. A final source state does not, by itself, specify lifted phase action, positive traversal time, or the response of a coupled field. We develop a source-resolved construction in the Substrate Accumulation Model (SAM) that keeps these quantities distinct. Original signed Write incidence yields an invertible three-phase map and the action one-form $\hbar(\mathrm{d}\gamma + 3\mathrm{d}\chi)$; certified H02–H06 histories demonstrate positive, order-dependent clock time. The full pair source has six driven two-state lanes and four unforced states. A subsequent original-Write source calculation admits four labelled source placements, constructs their finite boundary-incidence and signed-response matrices, and derives an explicit quartic source spectrum. The retained reference preparation has energy $E_{\mathrm{ref}} = 573049$, fixing a normalized receiving chart $\epsilon_R = 1/573049$ when the stated action and time conventions are selected; it does not fix the physical coupling without those assignments. Six signed lane couplings then enter the complete two-polarization quadrupole receiving action, with the full source energy, boundary interaction and both counterterm contributions retained. Source–field correlations give an exact actual-phase clock equation and conditional pointwise and finite-window bounds. A separate spin transfer computes the previously missing J_{21} and F_{11} source contractions for 54 preparations: all 1,998 five-coordinate upper enclosures improve while the completed lower remains fixed. The manuscript distinguishes source rules, receiving laws, mathematical compositions, and numerical qualifications; neither the spin upper nor the normalized clock chart is substituted for a derivation of universal physical field coupling.

Keywords: substrate Writes; six received clocks; interacting source–field action; retained history; certified spin reduction; relative entropy; source provenance.

1. Introduction

The motivating question is not whether two calculations can end at the same address. It is what remains physically and mathematically different when they do. A source can return to its wrapped phase endpoint while retaining a nonzero action increment. Two source paths can carry the same total signed phase and nevertheless require different positive elapsed times. A field coupled during those paths can preserve further differences in its state, energy, and correlations. The SAM research record considered here treats source state, executed history, clock readout, and receiving dynamics as separately retained objects [1, 4, 5].

The resulting construction has developed through successive uses of the same source information. Certified integration methods transferred from the frustrated-spin work made the original source clocks usable throughout five complete carrier trajectories. Later calculations retained the full mixed derivatives and inverse responses of those sources, rather than reducing them to endpoint values or scalar summaries. A signed endpoint-to-tensor map then fixed phase orientation and common-phase accounting. The supplemental derivation considered in this manuscript receives that work into a complete interacting pair-source and quadrupole-field action [1, 2, 4, 5, 6]. A subsequent source-spectrum calculation now identifies four admitted Write components and supplies the previously symbolic q_a values, a chart-dependent receiving normalization, and source-resolved six-lane clock relations [8]. Separately, the frustrated-spin continuation completes the missing J_{21} and F_{11} moments and a five-coordinate upper across the original 54 preparations [9, 10, 11].

Here *complete interacting action* has a precise model-level meaning: all terms in the specified joint source–field law are present, including the interaction expectation, both field polarizations, the source energy outside the current-rotation sector, and the counterterm. It does not mean that the quadrupole bulk law, quantization, autonomous incidence, and all physical unit assignments have been uniquely deduced from the original Writes. The supplemental source explicitly identifies those SAM_land receiving laws separately from the original Write rules [6].

The central contribution is therefore a connected mathematical formulation with a source and evidence record, not a replacement of all its inputs by a single asserted primitive. Sections 2–5 give the phase/source construction, received spectrum, interacting action, field normalization, and clock coefficients. Sections 6–8 relate them to completed history, radial-action and information-geometric calculations. Section 9 distinguishes the new six-clock source calculation and spin-method transfer from prior computational evidence. Detailed source matrices, Schur derivations, and the evidence register appear in the appendices.

1.1 Source layers and evidence convention

We distinguish three source layers throughout. *Original rules* supply signed endpoint Writes, source phase/action relations, retained tensor and carrier constructions, and the declared exterior metric. *Receiving laws* specify the SAM_land quadrupole bulk field, quantization, and autonomous source–boundary incidence. *Compositions* are the algebraic, geometric, and numerical consequences obtained after those inputs have been fixed. A receiving law does not become an original Write rule merely because the subsequent calculation is exact [2, 5, 6].

The numerical statements below are attributed to the supplied completed campaign reports. Their linked machine-readable receipts and derivation files are identified in Appendix F; they were not all included in the manuscript input package. No GEN6 execution was launched to write this manuscript. Newer source reports now describe native source-spectrum and hard-gap/finite-window clock calculations, while the completed spin continuation reports a distinct source integration and Duck receiver campaign [8, 11]. Their detailed raw interval tables and native execution manifests were not included among the four files supplied for this revision. We therefore reproduce their declared methods, numerical displays and source-reported outcomes, but do not invent absent exact endpoints or assign a fabricated total pass count.

1.2 Objects, units, and overloaded symbols

The finite pair source, the earlier four-by-eighteen Si29 source, the full18 tensor representation, and the full68 carrier are not interchangeable state spaces. They are connected only by the explicitly reported maps. Likewise, model time τ, the earlier source elapsed time t, a stationary phase $\Omega\tau$, and the actual accumulated phase γ are distinct coordinates. A reciprocal clock coefficient is meaningful only with its stated time and velocity convention.

We retain the source notation, adding descriptive subscripts where a collision would otherwise be ambiguous. Thus κ_{clk} denotes the earlier clock-curvature coefficient, while $\kappa = 4/5$ is the quadrupole field normalization. K_{pair} is the commuting remainder of the pair energy; the preparation-bound quantity written K in the latest supplement is not identified with it. R in the interacting action is the retained boundary radius, not the radix used elsewhere in SAM. Appendix G records these conventions.

2. Original Write incidence and the complete pair source

2.1 An invertible phase map

Let ϕ_a, ϕ_b, ϕ_p denote the lifted phases of the two endpoints and their partner. The original signed J4 actions determine the receiving coordinates [5]

$$\begin{aligned} \gamma &= \phi_a + \phi_b - 2\phi_p, \\ \delta &= \phi_a - \phi_b, \qquad \chi = \phi_p. \end{aligned} \tag{1}$$

The complete inverse is

$$\begin{aligned} \phi_a &= (\gamma + \delta)/2 + \chi, \\ \phi_b &= (\gamma - \delta)/2 + \chi, \\ \phi_p &= \chi. \end{aligned} \tag{2}$$

Both positive endpoint Writes induce the reported clockwise current quarter-turn; the positive partner Write receives the inverse paired action. With original tensor origin $(0, 2)$, the inherited counterclockwise physical-plane coordinate is

$$\psi = \psi_0 - \gamma/2. \tag{3}$$

The orientation sign and the half-angle factor belong to this source incidence. They are not selected after comparing a receiver output.

On the legal piecewise active Write paths, the original phase-action ledger becomes

$$\omega_{\mathrm{Write}} = \hbar(\mathrm{d}\gamma + 3\mathrm{d}\chi). \tag{4}$$

Substituting Eq. (2) gives $\mathrm{d}\phi_a + \mathrm{d}\phi_b + \mathrm{d}\phi_p = \mathrm{d}\gamma + 3\mathrm{d}\chi$. The coefficient three is therefore a coordinate-incidence coefficient. The separately retained common phase is part of the action account even when an observable receiving map is insensitive to it. This is an action relation on the stated Write paths; it is not by itself the entire source–field variational action [5].

2.2 Observable composition and its retained complement

The reported full18 tensor map has Gram matrix $32I_2$, while the full pair-observable map has Gram matrix $11062272I_2$. Native inverses compose these maps into a real/imaginary full18-to512 observable map. It reconstructs the two tensor coordinates, and the retained source complement reconstructs the remaining source coordinates. Full intertwiner, inverse, and leakage residual payments enclose zero in the recorded calculation [5].

The successful statement concerns this observable composition. The literal endpoint Hilbert action is different: even with the appropriate paired half-turn, both full D/C residual Grams retain exact $(0, 0)$ entry 54272. This nonzero result is preserved. An observable intertwiner is therefore not promoted into an equality of the complete Hilbert actions.

All sixteen joint C_4 character projectors are retained. Sector $(0, 0)$ has dimension three and each other sector dimension one, summing to the full18 identity. The tensor observation sees exactly sectors $(1, 1)$ and $(3, 3)$, with weights

$$\begin{aligned} W_{11} &= 16\begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}, \\ W_{33} &= 16\begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}, \\ W_{11} + W_{33} &= 32I_2. \end{aligned} \tag{5}$$

The two visible sectors do not remove the other source sectors; those remain in the complementary reconstruction. No logarithm branch was chosen for this decomposition [5].

2.3 Six driven lanes and four unforced states

The retained pair source is

$$\begin{aligned} H_{\mathrm{pair}} &= \mathrm{Swap} + Q_L + Q_R + Q_L Q_R, \\ D &= Q_L - Q_R, \qquad C = i\,\mathrm{Swap}\,D. \end{aligned} \tag{6}$$

Its sixteen-dimensional space comprises six driven two-state lanes and four unforced states. The earlier source result also retains

$$\begin{aligned} N_s &= (I + \mathrm{Swap})/2, \\ H_{\mathrm{pair}} &= 2N_s + K_{\mathrm{pair}}, \\ K_{\mathrm{pair}} &= E_{\mathrm{even}} - I, \qquad E_{\mathrm{even}} = Q_L + Q_R + Q_L Q_R. \end{aligned} \tag{7}$$

The remainder commutes with both currents. Consequently, a map that determines the rotation of D and C need not determine the whole source energy: K_{pair} and the six lane occupation weights still have to be retained [5, 6].

For a symbolic pair of source eigenvalues q_a, q_b, the lane spanned by $|ab\rangle, |ba\rangle$ makes this distinction transparent. With $d_{ab} = q_a - q_b$, the lane currents obey $D^2 = C^2 = d_{ab}^2 I$, while the even source contribution is $(q_a + q_b + q_a q_b)I$. The earlier source note left the q_a values symbolic. The new admitted four-placement calculation now supplies a conditional spectrum from which all six differences d_{ab} can be formed; the coupled lane weights and full even source energies remain explicitly accounted for, rather than inferred from the current rotations alone [8]. Appendix A expands the pair algebra without collapsing these types.

2.4 Original Write incidence and the four-source spectrum

The new source construction begins with labelled cube paths 46501, 46525, 49061 and 81829. Each admitted original single-address Write flips one of three address bits at one of seven occupied slots. The complete forbidden single-hop boundary consists of 78 distinct target addresses; its signed incidence matrix B has size 4×78. The native source product $T = \mathsf{B}\mathsf{B}^*$ is reported as [8]

$$T = \begin{pmatrix} 84 & 8 & 8 & 0 \\ 8 & 84 & 0 & 0 \\ 8 & 0 & 84 & 8 \\ 0 & 0 & 8 & 84 \end{pmatrix}. \tag{8}$$

The accompanying signed-response operand is assembled from the completed original source map as $K = \sum_{\text{axis}} D_{\text{axis}} T D_{\text{axis}}$ and takes the retained value

$$K = \begin{pmatrix} 672 & 96 & 32 & 0 \\ 96 & 2016 & 0 & 0 \\ 32 & 0 & 672 & 32 \\ 0 & 0 & 32 & 672 \end{pmatrix}. \tag{9}$$

The selected original response ledger was not present in the transmitted source record, so K is a completed retained operand, not a newly reconstructed unpublished ledger. The species and exchange prescriptions in its receiving map preserve their recorded exploratory provenance. Given the explicit receiving strength g_H, the source matrix is

$$Q(g_H) = T + g_H K. \tag{10}$$

Set $x = q - (84 + 672 g_H)$, $a = 8 + 96 g_H$, $b = 8 + 32 g_H$ and $h = 1344 g_H$. The exact characteristic equation is

$$\begin{aligned} 0 = x^4 - h x^3 - (a^2 + 2b^2) x^2 \\ + 2 h b^2 x + a^2 b^2. \end{aligned} \tag{11}$$

For the inherited *unit-strength receiving chart* $g_H = 1$, not an inferred physical coupling, four disjoint source roots are reported as in Table 1. Their exact characteristic coefficients, rational root brackets and interval-Newton certificates are distinguished from the rounded display values [8].

Table 1: Reported approximate eigenvalues of $Q(1)$ for the four admitted original Write components. Six off-diagonal differences d_{ab} follow without fitting the individual roots.

Source root	Displayed eigenvalue q_a
q_0	697.3304925912571
q_1	752.0081836576553
q_2	810.6543502862722
q_3	2108.0069734648155

2.5 Reference capacity and six signed receiving scales

The same full pair Hamiltonian of Eq. (6) retains the even energy $Q_L + Q_R + Q_L Q_R$ as well as exchange. For the original reference $|00\rangle$ in the unit-strength receiving chart, $Q_{00} = 756$, and the full reference energy is exactly

$$E_{\text{ref}} = (1 + Q_{00})^2 = (1 + 756)^2 = 573049. \tag{12}$$

The original role-capacity expression is $E_{\text{ref}}^{\text{phys}} = \hbar \lambda_{\text{ref}} / t_{\text{SW}}$. Equating that *same specified preparation* with the action-unit I_s, model-time unit t_u and scaled source Hamiltonian gives

$$\epsilon_R = \frac{\hbar}{I_s} \frac{t_u}{t_{\text{SW}}} \frac{\lambda_{\text{ref}}}{E_{\text{ref}}}. \tag{13}$$

In the explicitly normalized chart $I_s = \hbar$ and $t_u = t_{\text{SW}} / \lambda_{\text{ref}}$, this becomes $\epsilon_R = 1/573049$. It is a chart-level coefficient conditioned on the reference capacity and time conventions; no physical numerical value of t_{SW}, λ_{ref} or the universal interaction strength is thereby measured. The four q_a define six signed differences $d_{ab} = q_a - q_b$ and couplings $g_{ab} = \epsilon_R d_{ab}$; both their signs and the original even energies are retained [8].

The original active-Write rule also supplies a different, signed clock-current map,

$$\begin{aligned} \dot{\phi}_j &= s_j \frac{c(1 - A) D_j}{R_H}, \\ \dot{\gamma} &= \frac{c(1 - A)}{R_H} (s_a D_a + s_b D_b - 2 s_p D_p), \\ \epsilon_{\text{current}} &= \frac{t_u}{2} \dot{\gamma}. \end{aligned} \tag{14}$$

Here R_H is the source sphere, not automatically the quadrupole boundary radius R. A preparation-dependent signed phase rate cannot be identified with a positive autonomous receiving constant without a source incidence map. The common phase χ is separately retained in Eq. (4). The corresponding action-unit compatibility condition remains a stated test, not an imposed equality.

3. A joint interacting variational action

3.1 The complete joint-state expression

Let $|\Psi(\tau)\rangle$ be a normalized state of the full source tensor the complete field space. The supplemental derivation specifies the model-time action [6]

$$\begin{aligned} S = I_s \int \mathrm{d}\tau \Big\{ \frac{i}{2} \big(\langle \Psi | \dot{\Psi} \rangle - \langle \dot{\Psi} | \Psi \rangle \big) \\ - \langle \Psi | H_{\text{tot}} | \Psi \rangle \Big\}, \end{aligned} \tag{15}$$

with

$$H_{\mathrm{tot}} = \epsilon_R H_{\mathrm{pair}} + H_{\mathrm{free}} - \epsilon_R\big[D\phi_+(R) + C\phi_\times(R)\big] + \frac{\epsilon_R^2}{4R}(D^2 + C^2). \tag{16}$$

Here ϵ_R is the retained receiving scale, I_s is the action unit, and $\phi_+, \phi_\times$ are the two real field polarizations at the boundary. The new source calculation gives the conditional normalized chart $\epsilon_R = 1/573049$ through Eq. (13); the physical coupling still depends on the explicitly retained reference-role, action-unit and time assignments. It is not determined by equating the signed current rate with a universal source-energy coefficient [8].

The interaction remains a *joint* expectation. In particular, $\langle D\phi_+\rangle$ is not replaced by $\langle D\rangle\langle\phi_+\rangle$, and the corresponding cross-polarization correlation is retained. This distinction becomes essential in the actual phase-clock equation below. Source and field actions must not be added a second time to Eq. (15); it already contains the complete joint Hamiltonian.

Variation under normalization gives the full Schrödinger equation, with a scalar normalization multiplier contributing only a common state phase. This is the evolution of the specified source–field model. The original tensor metric, complete carrier, and continuous source construction provide the boundary incidence, whereas the SAM_land quadrupole bulk law, quantization, and autonomous incidence are explicitly the receiving laws [6].

3.2 Static normalization from both sides of the boundary

The literal transverse-traceless tensor metric is $B = 2I_2$. For one polarization, write the boundary coordinate as $x = R\phi(R)$. The retained regular interior and complete static exterior profiles are

$$\phi(r) = \begin{cases} xr^2/R^3, & 0 \le r \le R, \\ xR^2/r^3, & r \ge R. \end{cases} \tag{17}$$

Their gradient-plus-angular integrals are respectively

$$\int_0^R \big[r^2(\phi')^2 + 6\phi^2\big]\mathrm{d}r = 2x^2/R, \qquad \int_R^\infty \big[r^2(\phi')^2 + 6\phi^2\big]\mathrm{d}r = 3x^2/R. \tag{18}$$

Both pieces are needed. Matching their sum to the stated tensor metric gives, componentwise,

$$\frac{5\kappa}{4} = \frac{B}{2}, \qquad \kappa = \frac{2B}{5} = \frac{4}{5}. \tag{19}$$

In the final equality B denotes the value two on either real polarization. The radial coefficient is therefore $\kappa/4 = 1/5$. It has a different source meaning from the tensor-history one-form coefficient $1/4$ cited by the supplement from H002624; neither coefficient is substituted for the other [6].

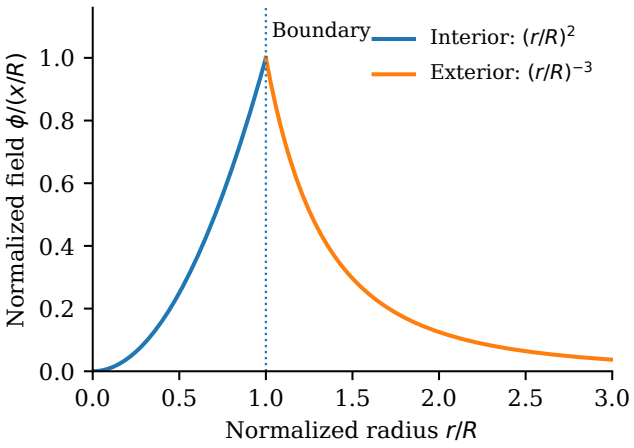


Figure 1: The retained static quadrupole profile, rendered directly from Eq. (17). The plotted normalization is $\phi/(x/R)$ against r/R. Interior and exterior pieces meet at the boundary, but their slopes are 2 and −3, respectively. This is an analytical illustration of the supplied profiles, not a new native field evolution.

3.3 Canonical field coefficients and the boundary jump

For the two real polarizations, the receiving free-field Lagrangian is

$$L_{\mathrm{free}} = \frac{\kappa}{4}\sum_{a=+,\times}\int_0^\infty \mathrm{d}r\,[r^2\dot\phi_a^2 - r^2(\partial_r\phi_a)^2 - 6\phi_a^2]. \tag{20}$$

With measure $r^2\mathrm{d}r$, the momentum is $p_a = (\kappa/2)\dot\phi_a$. With measure $\mathrm{d}r$, it is the density $\Pi_a = r^2 p_a$. These are two representations of the same canonical variable, not different momentum laws. The kinetic Hamiltonian density is p_a^2/κ with the first measure, or $\Pi_a^2/(\kappa r^2)$ with the second. The coefficient chain is

$$\frac{\kappa}{4} = \frac{1}{5}, \quad \frac{\kappa}{2} = \frac{2}{5}, \quad \frac{1}{\kappa} = \frac{5}{4}, \quad \frac{6\kappa}{4} = \frac{6}{5}. \tag{21}$$

The canonical action contains $+\epsilon_R J_a\phi_a(R)$ and the negative counterterm, where $J_+ = D$ and $J_\times = C$. Its stated field equation is

$$\frac{\kappa}{2}\big[r^2\ddot\phi_a - \partial_r(r^2\partial_r\phi_a) + 6\phi_a\big] = \epsilon_R J_a\delta(r - R). \tag{22}$$

The integrated equation retains the signed derivative jump

$$[\partial_r\phi_a]_{R^-}^{R^+} = -\frac{2\epsilon_R J_a}{\kappa R^2}. \tag{23}$$

Appendix B gives the explicit static integrals, Legendre transform, and jump calculation. Their status is a derivation from the receiving law, not a new numerical receipt.

3.4 Counterterm and positive field form

For a single current component J, the static boundary block completes exactly as

$$\frac{x^2}{R} - \frac{\epsilon_R J x}{R} + \frac{\epsilon_R^2 J^2}{4R} = \frac{(x - \epsilon_R J/2)^2}{R}. \tag{24}$$

The retained spectral density supplies

$$\mu = \int_0^\infty \frac{|u_R(\omega)|^2}{\omega} \mathrm{d}\omega = \frac{1}{5\kappa R} = \frac{1}{4R}. \tag{25}$$

Consequently, the field, interaction, and counterterm block of Eq. (16) is the sum over $a = +, \times$ of the positive forms

$$\int_0^\infty \omega \left(a_a - \frac{\epsilon_R J_a u_R^*}{\omega}\right)^\dagger \left(a_a - \frac{\epsilon_R J_a u_R^*}{\omega}\right) \mathrm{d}\omega. \tag{26}$$

The source energy $\epsilon_R H_{\mathrm{pair}}$ remains in the total Hamiltonian; the positive-form identity does not discard it. Nor does the identity assert that the two shifted operators are independent canonical annihilators. Their cross commutator retains the noncommuting source currents, so a simultaneous shifted vacuum is not inferred [6].

Within a driven lane, $g = \epsilon_R d_{ab}$ and $D^2 = C^2 = d_{ab}^2 I$. The total counterterm is therefore

$$2g^2\mu = \frac{g^2}{2R}. \tag{27}$$

Both contributions are paid. The even source energy, source excitation, complete free field, interaction, and counterterm are retained in the energy and action record.

3.5 Receiving all six source lanes into the continuum

The source eigenvalues in Table 1 label six pairs $a < b$ with $d_{ab} = q_a - q_b$ and signed $g_{ab} = \epsilon_R d_{ab}$. The quadrupole-field law, its quantization and the original retained boundary radius are *receiving assignments*, separately identifiable from the single-address Write incidence. For j_2 the order-two spherical Bessel function, the recorded spectral density and lane quantities are [8]

$$\begin{aligned} |u_R(\omega)|^2 &= \frac{2\omega j_2(\omega R)^2}{\kappa\pi}, \\ J_{ab}(\omega) &= 2g_{ab}^2 |u_R(\omega)|^2, \\ \mu &= \frac{1}{4R}, \qquad b = \frac{1}{6\kappa\pi}, \\ b_3 &= \frac{R}{42}, \qquad c_{ab} = \frac{g_{ab}^2}{2R}. \end{aligned} \tag{28}$$

The factor of two in J_{ab} retains both circular polarizations. The signed coupling g_{ab} remains in the operator interaction and phase correlations even when positive energy bounds contain g_{ab}^2. All four unforced states remain present in the sixteen-dimensional source sector. Neither the rotating-wave approximation nor a finite photon-number truncation is used to define the receiving law. Conditional physical time conversion multiplies a dimensionless angular rate by $\lambda_{\mathrm{ref}}/t_{\mathrm{SW}}$, rather than choosing an unsupported inverse-seconds value.

4. Both helicities and signed angular action

The source lowering combination is

$$L = \frac{D + iC}{\sqrt{2}}, \qquad [N_s, L] = -L. \tag{29}$$

The circular decomposition retains all four interaction words [6]

$$La_+^\dagger, \qquad L^\dagger a_+, \qquad L^\dagger a_-^\dagger, \qquad La_-. \tag{30}$$

Each conserves the signed charge

$$Q_{\mathrm{ch}} = N_s + N_+ - N_-, \tag{31}$$

which is the quantity denoted Q in the supplemental charge statement. The tensor angular action is $2I_s Q_{\mathrm{ch}}$.

The counter-rotating words change the unsigned source-plus-field excitation count by two, but they leave the signed charge unchanged. The law therefore retains both helicities and does not impose a rotating-wave truncation. The charge calculation uses the exact boson algebra and is not conditioned on a finite photon-number basis. Appendix A lists the charge changes of all four words.

The earlier phase-unit mapping states that receiving the same pair gap and clock under the original convention $E = \hbar\omega$ fixes $I_s = \hbar$, with the time conversion cancelling. This identifies the phase/action conventions; it is not a numerical measurement of an action unit or a determination of the full physical coupling scale [5].

5. Actual phase clocks and dependent action coefficients

5.1 The exact correlated phase equation

In a driven lane define

$$z = \frac{\langle D + iC\rangle}{d_{ab}}, \qquad \Delta = 2\epsilon_R, \qquad \gamma = -\arg z, \tag{32}$$

with the phase continued from its retained initial value. The supplemental action yields

$$\dot{z} = -i\Delta z + 2g\langle \mathrm{Swap}(\phi_\times - i\phi_+)\rangle. \tag{33}$$

The actual rate is

$$\nu_{\mathrm{actual}} = \dot{\gamma} = -\,\mathrm{Im}(\dot{z}/z), \tag{34}$$

where the amplitude is nonzero. The interaction term contains source–field correlations. Replacing it by products of marginal means would define a different clock equation [6].

For $A_a = \langle \mathrm{Swap}\,\phi_a\rangle$, the source and field commute and $\mathrm{Swap}^2 = I$. The supplied finite-energy form bound is

$$|A_a| \le 2\sqrt{\mu\langle H_{\mathrm{free},a}\rangle}. \tag{35}$$

It does not require an equal-time field variance. Combining both polarizations gives the stated instantaneous certificate

$$|\nu_{\mathrm{actual}} - \Delta| \le \frac{4|g|\sqrt{\mu E_{f,\mathrm{upper}}}}{m_{\mathrm{lower}}}, \qquad m_{\mathrm{lower}} > 0. \tag{36}$$

Here m_{lower} is the supplied positive amplitude lower bound. The bound is on an instantaneous rate, not on a fitted average frequency.

5.2 Stationary reference and finite-window phase bounds

The retained stationary hard-field gap is denoted Ω. It is distinct from both the bare gap Δ and the actual rate in Eq. (34). The all-order soft remainder and phase halfwidth received by the supplemental derivation are

$$\begin{aligned} \mathcal{E}(t) &= \exp(4g^2 v_{\mathrm{soft}} t^2) - 1, \\ m_{\mathrm{lower}}(t) &= M_{\mathrm{lower}} - \mathcal{E}(t), \\ a(t) &= \arcsin\big[\mathcal{E}(t)/M_{\mathrm{lower}}\big]. \end{aligned} \tag{37}$$

The bounds apply in the stated regime of positive amplitude margin and defined phase halfwidth. They enclose the actual phase as

$$|\gamma(t) - \Omega t| \le a(t). \tag{38}$$

Thus the finite-window mean rate lies in

$$\frac{\gamma(t)}{t} \in \left[\Omega_{\mathrm{lower}} - \frac{a(t)}{t},\ \Omega_{\mathrm{upper}} + \frac{a(t)}{t}\right]. \tag{39}$$

Between two windows $t_2 > t_1$, the corresponding bound is

$$\begin{aligned} \frac{\gamma(t_2) - \gamma(t_1)}{t_2 - t_1} \in \Big[&\Omega_{\mathrm{lower}} - \frac{a(t_1) + a(t_2)}{t_2 - t_1}, \\ &\Omega_{\mathrm{upper}} + \frac{a(t_1) + a(t_2)}{t_2 - t_1}\Big]. \end{aligned} \tag{40}$$

These statements use accumulated phase. Differentiating an envelope for γ would not supply a pointwise enclosure for $\dot\gamma$; the supplemental derivation does not make that substitution [6].

5.3 Clock conversion of the complete action

For the stationary reference $\gamma_h = \Omega\tau$, every energy term receives the same reciprocal Ω. Along the actual phase coordinate, every term receives the same reciprocal ν_{actual} wherever that rate is nonzero. In either case the conversion acts on the entire coupled expression, not on independent marginal extrema of its coefficients.

The velocity convention is essential. Write $\nu = \nu_{\mathrm{actual}}$ on an admissible phase segment. Then

$$\mathrm{d}\tau = \frac{\mathrm{d}\gamma}{\nu}, \qquad \dot\phi_\tau = \nu\, \partial_\gamma \phi. \tag{41}$$

Receiving a coefficient by $1/\nu$ while retaining $\dot\phi_\tau$ as its velocity is valid, but changing that velocity to $\partial_\gamma\phi$ introduces an additional ν^2. The phase-derivative coefficients are therefore

$$\begin{aligned} \text{field kinetic:} &\quad \kappa\nu/4, \\ \text{field gradient:} &\quad -\kappa/(4\nu), \\ \text{field angular:} &\quad -6\kappa/(4\nu), \\ \text{source:} &\quad -\epsilon_R/\nu, \\ \text{interaction:} &\quad +\epsilon_R/\nu, \\ \text{counterterm:} &\quad -\epsilon_R^2/(4R\nu). \end{aligned} \tag{42}$$

Every entry is evaluated with the same actual, joint-state-dependent rate. Their enclosures are those of a dependent coefficient family. The joint Berry one-form is unchanged under this reparametrization [6].

If a pointwise native rate enclosure contains zero, the specified response is to retain model time and the finite-window mean-rate inverse, not to assign a globally certified instantaneous inverse. A finite-window coefficient τ/γ also does not turn an arbitrary time-dependent monomial into a local phase density. For a term $H_a(s)$,

$$\frac{\int_0^\tau H_a(s)\mathrm{d}s}{\gamma} = \frac{\tau}{\gamma}\,\overline{H}_{a,\tau}. \tag{43}$$

The average on the right remains necessary. Conserved total energy is the stated exception: its dynamical energy action is $I_s E_{\mathrm{total}}\tau$, with action per accumulated phase coefficient $E_{\mathrm{total}}/(\gamma/\tau)$ when the finite-window lower rate is positive. This dynamical energy-action readout is distinct from the full variational action, which still contains its Berry term.

5.4 New receiving-bound family

The exact resolved-clock gzip outputs cited by the supplement were absent from its local receiving context. Rounded tables were not promoted to certified endpoints. Instead, the supplemental construction specifies new conservative bounds using the exact retained radius and six source lanes [6]:

$$\begin{aligned} c &= 2g^2\mu, \\ \mathcal{A} &= \frac{2g^2 b}{(1 - \Delta/\lambda)^2}, \qquad n_+ = \frac{\mathcal{A}}{1 + \mathcal{A}}, \\ M_{\mathrm{lower}} &= 1 - (1 + 2\mathcal{A})n_+. \end{aligned} \tag{44}$$

The symbol $\mathcal{A}$ is the preparation quantity written A in that bound, not SAM accumulation or enclosing area. With the exact soft-vacuum counterterm included, the received inequalities are

$$\begin{aligned} \langle K\rangle_{\mathrm{prep}} &\le c + \epsilon_R n_+, \\ \langle K\rangle_t &\le c + \epsilon_R[n_+ + \mathcal{E}(t)], \\ E_{f,\mathrm{upper}} &= (\sqrt{c} + \sqrt{K_{\mathrm{upper}}})^2. \end{aligned} \tag{45}$$

The supplement receives $b, \lambda, v_{\mathrm{soft}}$ and the preparation account K from its retained hard/soft construction. Their detailed underlying records and the new resulting numerical enclosures were not supplied for this draft. Equations (44)–(45) are therefore presented as the stated analytical receiving family. The retained tighter Gaussian certificates remain assigned to their original native root, rather than being replaced by these conservative formulas.

5.5 Six received clock preparations and source correlations

For each driven lane, the source receiving record keeps the completed equal-coherent hard charge-0/1 preparation and full soft vacuum. It selects $\lambda = 64\Delta$ with $\Delta = 2\epsilon_R$ in the normalization of Eq. (13). The original Gaussian trial, tails, branch isolation and occupation bounds are reported to enclose the true stationary hard gap Ω_{ab} and reference matrix element M_{ab}; detailed exact lane endpoints remain with the cited native graph rather than being reconstructed from rounded displays [8].

The correlation term in Eq. (33) is physically essential. For the stated initial preparation $z(0) = M_{ab} > 0$ and $\dot z(0) =$

Table 2: Completed H-series numerical successors [1]. Counts refer to the original carrier DAG and the reported independent comparison.

Case	DAG operations	Checks	Result
H02	957	16,192	PASS
H03	957	16,192	PASS
H04	953	16,172	PASS
H05	958	16,197	PASS
H06	956	16,187	PASS
Total	4,781	80,940	5/5

$-i\Omega_{ab}M_{ab}$, so

$$\begin{aligned} \nu_{ab}(0) &= \Omega_{ab}, \\ \langle \mathrm{Swap}\,\phi_{\times}\rangle_0 &= 0, \\ \langle \mathrm{Swap}\,\phi_{+}\rangle_0 &= -\frac{(\Delta-\Omega_{ab})M_{ab}}{2g_{ab}}. \end{aligned} \tag{46}$$

Thus the initial actual-phase rate is shifted from the bare gap unless the received correlation vanishes. The result is not reproduced by multiplying independent source and field means. For subsequent times, the common hard gap and the all-order soft envelope bound accumulated phase and its finite-window secant rate. They do not provide a pointwise rate by differentiation of the envelope. A separate independent checker is reported to use a distinct address reconstruction, exact Sturm root counting and full hypergeometric Gaussian expressions [8]. No fabricated six-row numerical certificate table is inserted here: the supplied derivation references, but does not reproduce, its raw certified endpoint intervals.

6. Completed history and clock constructions

6.1 Certified integration of H02–H06

The original frozen five-case experiment completed with zero full trajectories, and all five failures were independently reproduced. A separate numerical successor refined only the two incoming elapsed-duration enclosures for each case. The same source density was evaluated in angular and half-angle coordinates at 384 bits; every refined enclosure had to lie inside its original enclosure. Receiver precision remained 192 bits, logarithm tolerance remained 10^{-35}, and the maximum remained 512 terms. Original source events, waiting gaps, coefficients, initial states, and interpretations were unchanged [1].

All five successors completed their original carrier jobs. The record includes 4,781 original directed-acyclic-graph operations, all thirty required clock calls to `GEN5_LOG_QUADRATIC_FORM`, and 80,940 independent-verification checks. Table 2 preserves the case-level counts. The result is a method transfer that resolves the original inputs more tightly, not a relaxation of the acceptance criterion and not a relabelling of the frozen outcomes.

The independent route reconstructs the signed source paths by 120-digit angular quadrature, then uses the retained `FullModel` implementation for the 273/546-dimensional evolution. Native intervals and outputs do not enter that independent solver; comparison occurs after reconstruction. The shared model implementation is part of its stated scope, not concealed as a wholly different physical model. All six carrier segments, four paid/image events, source operators, clock rows, memory, outgoing and driver accounts, and full energy records are compared [1].

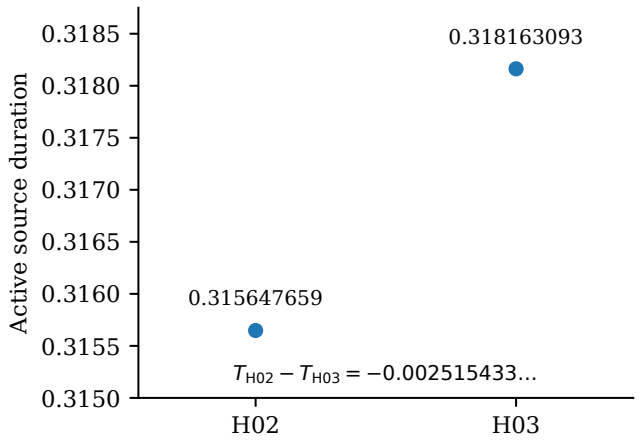


Figure 2: Reported active source durations for H02 and H03, at the unchanged original calibration [1, 4]. The vertical scale is expanded to display the ordering contrast. This plot does not combine these values with the distinct final receiver-clock readings and does not depict uncertainty widths.

6.2 Same endpoint, different positive durations

The H02 and H03 histories have the same source endpoint and zero total signed phase action. Their positive source durations are

$$\begin{aligned} T_{\mathrm{H02}} &= 0.3156476594819614\ldots, \\ T_{\mathrm{H03}} &= 0.3181630928556934\ldots, \\ T_{\mathrm{H02}} - T_{\mathrm{H03}} &= -0.002515433373731972\ldots. \end{aligned} \tag{47}$$

The negative last line is a difference of positive durations. The reported calibration remains $R_H = 1$, $c = 10$, $A_{\mathrm{los}} = 0$, elapsed scale one, and zero waits for these two histories [4]. Figure 2 displays only the source durations.

At their common internal receiver endpoint 0.35, the separately reconstructed final receiver times are approximately 1.5928865558203955 and 1.5928814590148197. Their ordering is not the same observable as the ordering of source durations. The final native intervals are disjoint in 81 of 273 current coordinates and 162 of 546 first-return coordinates. H04 returns to its wrapped source address with cancelling signed phase action, but retains positive traversal, paid work, and disjoint initial/final receiver intervals in 85 of 273 coordinates. H05 and H06 preserve their distinct waits $1/200$ and $1/250$ outside active traversal [1].

6.3 Local clock curvature and inverse response

On the original phase plane $\theta = (x, -y, 0)$, the retained source laws give [4]

$$\begin{aligned} E &= 6 - 2\cos x + 2\cos y, \\ q_a &= 2\sin x, \qquad q_b = 2\sin y, \\ F(E,q) &= \frac{12\sqrt{E^2+q^2}}{E^2+36}, \\ \alpha_{\text{clock}} &= \frac{F_a \mathrm{d}x + F_b \mathrm{d}y}{10}, \end{aligned} \tag{48}$$

where $F_a = F(E, q_a)$ and $F_b = F(E, q_b)$. The exterior derivative is

$$\begin{aligned} \mathrm{d}\alpha_{\text{clock}} &= \frac{\kappa_{\text{clk}}}{10}\mathrm{d}x \wedge \mathrm{d}y, \\ \kappa_{\text{clk}} &= q_a F_E(E, q_b) + q_b F_E(E, q_a). \end{aligned} \tag{49}$$

The lifted clock directions $X = \partial_x + (F_a/10)\partial_t$ and $Y = \partial_y + (F_b/10)\partial_t$ consequently obey

$$[X, Y] = \frac{\kappa_{\text{clk}}}{10}\partial_t. \tag{50}$$

The boundary ordering contrast is the integral of this local coefficient with the retained orientation. The $1/10$ is inherited calibration, not a newly derived unit conversion.

Opposite certified signs along the supplied half-angle path selected an adaptive zero bracket,

$$\frac{117}{128} < s_{\text{zero}} < \frac{59}{64}, \qquad \text{width} = \frac{1}{128}. \tag{51}$$

The record establishes at least one zero by continuity, not uniqueness. Its next midpoint was prepared but unexecuted after the admission window. Full four-coordinate inverse closures and signed mixed role-metric terms also passed: full geometric payments were approximately 1.999472685543 and 2.000665760180, compared with diagonal portions 2.186193870186 and 2.244401690354 [4].

6.4 Lifted action and wrapped collisions

The later phase-character calculation groups 64 wrapped source addresses into 32 pairs. Appending lifted displacement $(2, 2, 0)$ is invisible to the three wrapped receiving characters but adds four quarter-action units. A four-W1 word returns to the same source endpoint while retaining winding and action increment $2\pi\hbar$. All 65 lifted inverses and wrapped-collision checks close exactly in the report [5].

These inverses recover cumulative lifted phases. The literal event order remains in the retained words, every prefix, and positive clock history; it is not reconstructed from cumulative phases alone. This separates two source mechanisms: a wrapped observation may lose lifted action, while a clock one-form may retain path ordering even when total signed phase vanishes. Both mechanisms are kept when the source is received into the interacting law.

6.5 Finite retained-history inverse and composition control

An independently scoped six-history Focused campaign provides a useful computational control rather than a new physical clock law. Its E/F histories share their complete whole-history scalar tuple (U, D, V, L, M) and their step-ratio and directed-transition multisets, yet their retained internal range $[1, 4)$ has net logarithmic change $+\ln 2$ and $-\ln 2$, respectively. Access to one declared chronological coordinate reduces each of the six original two-candidate inverse fibers to its original single member. The native composition reuses retained chunks without resummarizing their edges, and a fresh Wolfram evaluation agrees on 48/48 selected subrange/inverse comparisons [12].

This engineering experiment also preserved 75 completed and 14 failed *top-level wrapper* task outcomes. Thirteen method-status failures were recorded despite underlying admitted, compiled and passed native receipts; recovery used typed object references rather than relabelling the failed wrapper calls. The demonstration is exact finite-history information and execution reuse. No physical duration, source action or coupling is assigned to its synthetic state values.

7. Radial action and full coordinate response

An earlier completed cycle composes the declared static nonrotating exterior metric with the original worldline action $S = -E_0\tau$. With $f = 1 - A$, radial ruler $\mathrm{d}r/\sqrt{f}$, and static clock $\sqrt{f}\,\mathrm{d}t$, its result is [2]

$$L_{\text{rad}} = -E_0\sqrt{(1-A) - \frac{\dot{r}^2}{c^2(1-A)}}. \tag{52}$$

The fixed-role timelike domain is $|\dot{r}/c| < 1 - A$. G471 supplies the role energy $E_0 = \hbar\epsilon_{\text{role}}/t_{\text{SW}}$, with its original source expression for ϵ_{role}. The weak-field terms are $-E_0 + E_0 A/2 + mv^2/2 + \cdots$. This is a composition with an original *declared* metric, not a fresh derivation of that metric from microscopic Writes.

At $f = 9/16$ and $\beta = \dot{r}/c = \pm 27/80$, the proper-time lapse is $3/5$ and the normalized Hessian is

$$K_{\text{rad}} = \begin{pmatrix} 433/135 & \pm 11/3 \\ \pm 11/3 & 125/27 \end{pmatrix}. \tag{53}$$

Full Schur elimination gives

$$\frac{433}{135} - \frac{(11/3)^2}{125/27} = \frac{1024}{3375}, \tag{54}$$

matching the independently calculated fixed-momentum Routh curvature. The conditional inverse response retains opposite signs under velocity reversal.

With local speed $\zeta = \beta/f$, the full nonlinear coordinate pullback gives mixed entry $-1/2$ at positive speed, while its Jacobian-only part gives $+1/2$. Their diagonals, determinant, and spectrum agree. The full tensor, not a spectral summary, identifies the different response. Likewise, rank-three source chart metrics retain distinct inverse phase norms even at addresses with zero contact Hessian [2].

The earlier clock model with inertia $1 + A$ and stiffness one has frequency squared $1/(1 + A)$, not $1 - A$. Keeping that inertia would conditionally require stiffness $1 - A^2$. The separate proper-time conversion $\mathrm{d}\tau = \ell\,\mathrm{d}t$ instead gives $M_t = M_\tau/\ell$ and

$K_t = \ell K_\tau$. These conditional mappings motivate the explicit velocity bookkeeping in Eq. (42); they do not independently assign the microscopic clock stiffness [2].

8. Relative entropy, area, and reconstructible source coordinates

The Si29 comparison uses the retained four-by-eighteen tensor source, seed $(3, 10)$, original Q/G matrices, reference phases $7/1000$ and $13/1000$, local phase perturbation u, and interaction-phase perturbation v. Its area is the existing common-energy $e = 4$ enclosing observable $\bar{A} = A/(4\pi a^2)$ [3]. It is not identified with a local Rindler horizon cross section.

Across 34 signed perturbations, the calculation retains the decomposition

$$\mathcal{D}(\rho\|\sigma) = \Delta\langle K_\sigma\rangle - \Delta S_{\mathrm{vN}}. \tag{55}$$

Local u motion preserves entropy, whereas v changes the reduced spectrum. Relative entropy has zero first derivative at each reference, while $\bar{A}$ has a nonzero first derivative. Therefore the tested fixed-coefficient relation $\Delta\bar{A} = c\mathcal{D}$ fails for that area and those source motions. This is the exact scope of the report's classification, "The test falsifies the concept" [3].

The same nonzero geometric response supplies a useful alternative. For $x = (u, v)$ and $y = (S_{\mathrm{vN}}, \bar{A})$,

$$J = \begin{pmatrix} 0 & S_v \\ \bar{A}_u & \bar{A}_v \end{pmatrix}. \tag{56}$$

The reported determinant enclosures exclude zero at both references, with values approximately 3.36360099694 and 3.55314285468. Entropy locally resolves the interaction phase and area supplies the additional source orientation. The inverse is differential, not global: at step 10^{-4} the largest u errors are 7.8844×10^{-5} and 1.0231×10^{-5}; halving the step reduces them to 1.9675×10^{-5} and 2.5604×10^{-6} [3].

With the complete relative-entropy Hessian G and $B_{\mathrm{inv}} = J^{-1}$, the transported information metric is

$$G_y = B_{\mathrm{inv}}^T G B_{\mathrm{inv}}. \tag{57}$$

Its mixed entries remain nonzero. The original tensor Hamiltonian supplies trajectory tangent $w = (1 + 8\nu_{\mathrm{Si}}, \nu_{\mathrm{Si}})$ in the stated frame, giving $g_{tt} = w^T G w$. The resulting source-derived information coefficients are retained in Appendix E. They are not automatically physical action or clock coefficients: the original clock, field coupling, and boundary dynamical assignments must still connect the quantities [3].

Dorau and Much's supplied paper provides the comparison setting [7]. Their argument links relative entropy for coherent excitations to horizon energy flux, assumes the proportionality to horizon-area variation, and uses the quarter-area relation to fix the coefficient of the semiclassical Einstein equations. The finite Si29 enclosing area, perturbation family, and entropy ordering remain distinct. The failed direct proportionality in Si29 is not a test of their horizon claim. Its productive result is a two-observable local reconstruction and a full information tensor available for a future source-resolved boundary-response comparison.

9. Native execution, independent reconstruction, and reuse

9.1 What was executed

The completed reports separate native mathematical execution from local source retrieval, and from independent reconstruction. The radial-action campaign completed 82 mathematical tasks and nine retrieval tasks, with 2,485 native operations, 5,418 native checks, and 25,447 independent mathematical/provenance checks. The source-clock campaign completed 17 mathematical and eight retrieval tasks, with 709 native operations, 1,422 native checks, and 4,360 independent reconstruction checks. The tensor-channel campaign completed ten mathematical and seven retrieval tasks, with 373 native operations, 746 native checks, and 48,226 independent checks [2, 4, 5].

Each of those reported Research cycles retained two 300-second admission windows, outcome-selected second-round work, completed custody, and resource release. These timings bound admission, not a universal wall-clock promise for all derivation or verification work. The Si29 calculation is different: it is a focused warm-session native follow-up, not another timed Duck cycle. Its combined result is 1,116 successful native operations, two checkpoints, and 959 independent checks [3].

The earlier H-series and radial/Si29 results use their recorded core 0.10.1 identities. Later clock and tensor cycles report Duck 1.2.2 and integration core 0.10.2, with 716 Duck files and 743 shared native/library files matched at launch, runtime, and final custody. These are distinct recorded execution identities; the manuscript does not silently backdate the later version into earlier work [1, 2, 3, 4, 5].

9.2 Reconstruction and interval discipline

Independent verification is described at the level of the actual method, not by a single uniform label. The H-series route reconstructs source integrals independently but uses the unchanged retained carrier model. The radial-action checker derives derivatives using symbolic and exact rational algebra, verifies native traces and causal bindings, and bounds retained transcendental contractions. The later clock checker uses symbolic differentiation, exact containment, and 150-digit quadrature and matrix reconstruction. The tensor checker uses original matrices and exact rational, complex, and quadratic algebra [1, 2, 4, 5].

Si29's independent route uses 100-digit arithmetic, separate four-state/eighteen-state eigensystems, and independently solved radius roots. Native density intervals remain intervals; an exact-entry entropy convenience interface was not used to silently collapse them. The calculation instead composes entropy from existing logarithm-trace and matrix operations. Its first retrieval attempt failed on historical floating-point metadata and produced no mathematical result; the corrected exact-source route and the failed attempt both remain recorded [3].

These practices support reproducible comparison within the specified models. They do not convert a high-precision numerical reconstruction into an interval proof where the checker supplied only numerical values, nor does a predicate count measure the independence of distinct mathematical claims. The source reports identify what each check actually examines.

9.3 Retained results are not assumed to be faster

Reuse has several roles: retaining exact source identity, carrying a completed object into another operation, and avoiding repeated large constructions. The radial-action cycle reused complete-source contractions from a 4,194,304-state ordered-carrier propagation rather than rerunning it. The Si29 inverse/metric follow-up used thirty native operations and one checkpoint without recomputing forward states [2, 3].

A separate small-query comparison nevertheless found fresh integration faster than retained-operand admission. Three interleaved observations per route gave 3182.813 verified queries per CPU-hour for the retained route and 4483.138 for fresh certified integration. Both returned the same four clock observables under the common enclosure and source-overlap checks, while retaining different evidence. Process/operating-system caches were warm, native sessions were fresh, and historical construction costs were not amortized. These are internal method measurements, not an external throughput or dollar comparison [4].

The research use of retained information is therefore not reduced to caching. Its demonstrated value includes source-preserving composition, inverse reconstruction, and the retention of mixed terms and complementary state information that a cheaper scalar summary can omit.

9.4 An admitted four-source spectrum and certified six-clock route

The source-to-clock continuation now supplies $T = \mathsf{BB}^*$ from four actual cube paths and 78 forbidden single-hop targets, retains the signed source response K, and receives $Q(g_H)$ without fitting four eigenvalues. At the specified $g_H = 1$ chart the characteristic polynomial agrees algebraically with the four displayed eigenvalues. The source reference $E_{\mathrm{ref}} = 573049$ and Eq. (13) give the chart normalization and signed six-lane couplings. The full continuum construction then receives hard stationary gaps and finite-window phase enclosures in the conditional model time, with the full field, both helicities and counterterm included [8]. This is a later source-resolved continuation of the interacting action, not a retroactive claim that the quadrupole law or g_H has been deduced from primitive Writes.

The supplied new derivation names exact Sturm, rational-bracket, interval-Newton and independent hypergeometric reconstruction routes. Those are source-reported qualification methods; the complete native lane result table and immutable receipts were not uploaded with this manuscript update. We record neither invented numerical widths nor an invented total verification count. The new preparation-specific source assignment is nonetheless a substantive advance over a wholly symbolic source-spectrum interface.

9.5 Frustrated-spin transfer: original J21/F11 source completion

A separate GEN6 investigation closes the previously missing fifth-coordinate upper of the finite frustrated-spin program. It derives the original source contractions $J_{21} = \langle w_e, A_{\mathrm{spin}} w_d \rangle$ and $F_{11} = \langle A_{\mathrm{spin}} w_d, A_{\mathrm{spin}} w_d \rangle$, where $A_{\mathrm{spin}} = N(I - P)$, for all 54 retained preparations. The source adapter enumerates all $1,048,576$ core states per finite core and retains complete update fields, signed cross terms, Arb rate caches and Chebyshev moments. These source integrations were completed before the two GEN6 exploration windows; they are not counted as work performed during those windows [9, 10, 11].

Table 3: Reported normalized upper intervals across 54 spin preparations at $1000N$ site attempts. The lower L_5 is unchanged; these are ranges of upper values, not a measured distribution of gaps.

Upper family	Minimum	Maximum
Retained U_4	0.01735396	0.01739903
Source-completed U_5	0.01105964	0.01108194

Each of 54 original preparations contributes 37 retained histories, yielding $1,998$ improved upper enclosures, all strictly below their prior retained uppers. At $1000N$ original site attempts, the source report gives the normalized upper ranges in Table 3; the original five-coordinate lower L_5 is held fixed. The reported gain removes 54.76884–54.77327% of the *previously unresolved* $U_4 - L_5$ *gap*, not 54.77% of the raw upper value. The independent direct 5×5 receiver and scalar Schur receiver overlap in all 222 pilot comparisons. The adopted Schur denominator and gain are strictly positive across all original $1,998$ cases [11].

The source-action diagnostic further separates a direction factor close to unity from an independently measured source-action fraction near 0.5477 in the six pilots. The retained source-action excess F_{excess} lies approximately between 1.4209363 and 1.4212459. The proposed sixth-coordinate continuation would require a new action contraction and squared-action norm; it has not been executed by this experiment. This is a *method-transfer and source-action diagnostic*, not evidence that spin response coefficients are the universal source–field coupling constants of SAM [11].

The reported Duck 1.2.2/core 0.10.2 Connections cycle returned 84 mathematical and two indexed-source outcomes, performing $23,742$ native steps without failed mathematical returns. Its independent program made $1,764$ literal two-node core-rate/transition comparisons and $11,214$ interval-entry checks at 400 digits. Initial orchestration failures and subsequent repairs are preserved separately; run completion and resource release are reported, not independently re-executed here. Appendix I expands the receiver algebra and its exact provenance boundary.

10. Discussion and conclusion

The completed calculations identify several distinct ways in which a reduced observation loses information. Wrapped phase characters can erase lifted action. A final source endpoint and total signed phase can omit positive clock ordering. Eigenvalues can conceal the sign of a mixed coordinate response. Current rotation can omit a commuting source-energy contribution. Entropy can miss an orientation that is locally recovered by a geometric readout. These are specific mathematical statements with different mechanisms, not a claim that every history observable is interchangeable [1, 2, 3, 4, 5].

The interacting action receives that lesson in its structure. It retains a full joint source–field expectation, both helicities, all four circular interaction words, both counterterm contributions, and a phase clock whose exact rate contains source–field correlations. The field kinetic, gradient, angular, source, interaction, and counterterm coefficients change together under the same actual time conversion. Neither an independently selected clock rate nor a dropped mixed term can stand in for that dependent family [6].

Within the stated receiving model, the field normalization, Legendre transform, boundary jump, square completion, signed charge, and clock-conversion formulas form an explicit derivation chain. That is the constructive result of the supplemental action. The later source extension has now supplied an explicit finite source spectrum, a reference-energy receiving scale and a reported six-lane hard-gap/finite-window calculation. The exact source matrices and algebra are reproduced here. The detailed lane certificate endpoints, job hashes and full receiver check totals remain with the qualified project artifacts; they were not supplied in the update package and are not reverse-engineered from displays [8].

The original-source question also remains precisely located. The pair and Write inputs have not been conflated with the SAM_land quadrupole bulk and quantization laws that receive them. Establishing a physical source derivation of all those receiving assignments is a further task, rather than an unstated premise of this manuscript. Similarly, the Si29 information metric is a constructed comparison object, not an already identified gravitational metric.

We conclude that the supplied record now supports an interacting source–field formulation with a finite admitted Write spectrum, six signed lane couplings, a capacity-normalized receiving chart, explicit quadrupole boundary normalization, exact charge accounting, and state-dependent phase-clock relations. The completed histories, five H-series trajectories, source-action inverses and the independent Si29 construction form distinct supporting evidence. The five-coordinate spin upper establishes a separate source-bound benefit from retaining complete Gram payments and mixed terms. The combined advance is a more explicit map of *which* coefficients follow from original incidence, *which* are conditional receiving assignments, and *which* remain physical source-derivation tasks.

Contributions and acknowledgments

Sean Brady is the originator and conceptual director of SAM and the research program assembled here. OpenAI ChatGPT and Codex are AI research collaborators, technical contributors, and co-authors of the project research record. The supplied reports attribute source compositions, native programs, comparison design, and outcome-selected analytical interpretation to Codex. ChatGPT prepared this manuscript synthesis from the supplied material. GEN6 Duck, SLC, and the named domain engines are identified as computational systems rather than human authors. The journal-style byline in this draft is Sean Brady; no submission or institutional affiliation is implied.

Data, code, and version statement

This draft is based on the six supplied research records and the supplied Dorau–Much paper listed below. The internal reports identify retained code snapshots, source hashes, execution manifests, native intervals, failed attempts, and independent-verification records. Those links are provenance pointers, not a claim that a complete executable archive accompanies this manuscript. The companion manuscript-source package includes the article source, figure-generating script, supplied report copies where available, and a source-use register. The underlying research repositories and T500 stores are not modified. Numerical entries printed with decimals are reported approximations unless explicitly stated to be exact; rounded endpoints are not substituted for certified native data.

References

[1] SAM/GEN6 research record, *Spin integration across the original withheld cases*, 14 October 2026. Author-supplied report, file `Pastedtext(20261014-123442).txt`. Includes H02–H06 numerical successors, source refinement, independent reconstruction, and custody pointers.

[2] SAM/GEN6 research record, *GEN6 Duck: source action, phase, metric and ordered channels*, 14 October 2026. Author-supplied report, file `Pastedtext(20261014-132311).txt`; source directory `GEN4/duck_source_action1_20261014/`.

[3] SAM/GEN6 research record, *Si29 relative entropy, geometric response and reusable source coordinates*, 14 October 2026. Author-supplied report, file `Pastedtext(20261014-135802).txt`. Includes the focused native calculation and inverse/metric follow-up.

[4] SAM/GEN6 research record, *Original Write action: retained clock ordering and inverse coefficients*, 14 October 2026. Author-supplied campaign text; clock source reused in the subsequent report under H002622.

[5] SAM/GEN6 research record, *Original Write phases: tensor channels, inverse history and clock coefficients*, 14 October 2026. Author-supplied completed-cycle text, with full source-character, complement, lifted-history, and clock-receiving results.

[6] SAM/GEN6 supplemental derivation, *Complete interacting action*, supplied 14 October 2026. Receives the completed H002624 source-sector and six-lane work. New numerical results and native receiving receipts are deferred in the supplied version.

[7] P. Dorau and A. Much, "From Quantum Relative Entropy to the Semiclassical Einstein Equations," *Physical Review Letters* **136**, 091602 (2026), doi: 10.1103/lmq8-nsty. Comparison based on the supplied six-page published article.

[8] SAM/GEN6, *From Original Write Incidence to Six Received Clocks*, supplied source derivation dated 15 October 2026, `DERIVATION(4).md`. Four original Write paths,

source Gram/response matrices, quartic eigenvalues, E_{ref} and conditional clock receiving laws; raw exact native lane endpoints are referenced but not included in the supplied file.

[9] SAM/GEN6, *Scalar Completion of the Fifth-Coordinate Upper Response*, continuation of H002611 and H002626, source derivation supplied 15 October 2026, `DERIVATION (2).md`. Formulaic fifth-coordinate Schur reduction and Gram admissibility bound.

[10] SAM/GEN6, *Original-Source Completion of T5 and the Upper Response*, 15 October 2026, `DERIVATION(3).md`. Original J_{21}, F_{11} receiving route and independent direct/scalar formulas.

[11] SAM/GEN6, *Original J21/F11 Source Integrals and Completed Five-Coordinate Spin Upper*, owner-supplied project report dated 15 October 2026. Declares $54/54$ source-pair completions, $1,998$ improved uppers, pilot direct/scalar comparison and source-action diagnostics; native exact endpoints remain at the cited T500 source paths.

[12] SAM/GEN6, *Retained Histories Unlock Composition, Reuse and Finite Inverse Reconstruction*, author-supplied final supplement, 15 October 2026, `Pastedtext(2026 1015-165409).txt`. Native history controls, six finite inverse fibers, $48/48$ Wolfram comparisons, and preserved wrapper status failures.

A. Full derivation chain: phase incidence and pair algebra

A.1 Incidence, inverse, and action

Writing $\Phi = (\phi_a, \phi_b, \phi_p)^T$ and $\Theta = (\gamma, \delta, \chi)^T$, the supplied phase map and its inverse are

$$\Theta = M\Phi, \qquad M = \begin{pmatrix} 1 & 1 & -2 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad M^{-1} = \begin{pmatrix} 1/2 & 1/2 & 1 \\ 1/2 & -1/2 & 1 \\ 0 & 0 & 1 \end{pmatrix}. \tag{58}$$

Consequently,

$$\hbar(\mathrm{d}\phi_a + \mathrm{d}\phi_b + \mathrm{d}\phi_p) = \hbar\left[\frac{\mathrm{d}\gamma + \mathrm{d}\delta}{2} + \mathrm{d}\chi + \frac{\mathrm{d}\gamma - \mathrm{d}\delta}{2} + \mathrm{d}\chi + \mathrm{d}\chi\right] = \hbar(\mathrm{d}\gamma + 3\mathrm{d}\chi). \tag{59}$$

This expands the relation reported in Ref. [5]. The inverse acts on lifted cumulative phases; it does not reconstruct the ordering of an arbitrary word. Ordered words and clock history remain separately stored. The reported wrapped collisions are therefore consistent with an exact lifted inverse.

A.2 Algebra within one source lane

On the ordered basis $|ab\rangle, |ba\rangle$ with $a \neq b$, the supplied operators specialize to

$$\mathrm{Swap} = \sigma_x, \qquad D = d_{ab}\sigma_z, \qquad C = i\,\sigma_x d_{ab}\sigma_z = d_{ab}\sigma_y, \qquad H_{\mathrm{pair}} = \sigma_x + (q_a + q_b + q_a q_b)I. \tag{60}$$

This is an algebraic unpacking of Eq. (6), not a numerical assignment of q_a, q_b. It yields the stated equal current squares and identifies why the even source energy is invisible to the current rotation. With $N_s = (I + \sigma_x)/2$,

$$\left[N_s, \frac{d_{ab}}{\sqrt{2}}(\sigma_z + i\sigma_y)\right] = -\frac{d_{ab}}{\sqrt{2}}(\sigma_z + i\sigma_y) = -L. \tag{61}$$

The same pair algebra gives $[D, C] = -2i\,\mathrm{Swap}\,D^2$. The shifted polarization operators in Eq. (26) therefore retain a source-current cross commutator. Completing each square does not produce two independent oscillator vacua.

Table 4: Charge bookkeeping of the four retained circular interaction words. The changes sum to zero in $Q_{\mathrm{ch}} = N_s + N_+ - N_-$. The table expands the charge statement of Ref. [6]; it does not impose a photon-number truncation.

Word	ΔN_s	ΔN_+	ΔN_-	ΔQ_{ch}
$La_+^\dagger$	-1	$+1$	0	0
$L^\dagger a_+$	$+1$	-1	0	0
$L^\dagger a_-^\dagger$	$+1$	0	$+1$	0
La_-	-1	0	-1	0

B. Full derivation chain: field normalization, Legendre transform, and interaction

B.1 Interior and exterior payments

For the interior field $\phi = xr^2/R^3$, $\phi' = 2xr/R^3$. For the exterior field $\phi = xR^2/r^3$, $\phi' = -3xR^2/r^4$. Substitution into the supplied static quadratic form gives

$$\begin{aligned} \int_0^R [r^2(\phi')^2 + 6\phi^2]\mathrm{d}r &= \int_0^R \frac{10x^2r^4}{R^6}\mathrm{d}r = \frac{2x^2}{R}, \\ \int_R^\infty [r^2(\phi')^2 + 6\phi^2]\mathrm{d}r &= \int_R^\infty \frac{15x^2R^4}{r^6}\mathrm{d}r = \frac{3x^2}{R}. \end{aligned} \tag{62}$$

Thus one polarization contributes $5\kappa x^2/(4R)$. With the literal per-polarization metric value $B = 2$, matching to $(B/2)x^2/R$ gives $\kappa = 4/5$. This makes the static free energy x^2/R before interaction and counterterm are added. The calculation requires the complete exterior contribution; an interior-only profile would not supply this coefficient.

B.2 Canonical momentum in the two measures

Using $r^2\mathrm{d}r$ as the measure, the Lagrangian density and momentum are

$$\mathcal{L}_{r^2\mathrm{d}r} = \frac{\kappa}{4}\left[\dot\phi^2 - (\phi')^2 - \frac{6\phi^2}{r^2}\right], \qquad p = \frac{\partial\mathcal{L}_{r^2\mathrm{d}r}}{\partial\dot\phi} = \frac{\kappa}{2}\dot\phi. \tag{63}$$

The Legendre transform yields

$$\mathcal{H}_{r^2\mathrm{d}r} = p\dot{\phi} - \mathcal{L}_{r^2\mathrm{d}r} = \frac{p^2}{\kappa} + \frac{\kappa}{4}\left[(\phi')^2 + \frac{6\phi^2}{r^2}\right]. \tag{64}$$

With $\Pi = r^2 p$ and measure $\mathrm{d}r$, the identical Hamiltonian is

$$H_{\mathrm{free}} = \sum_a \int_0^\infty \mathrm{d}r \left[\frac{\Pi_a^2}{\kappa r^2} + \frac{\kappa}{4}\left(r^2(\partial_r\phi_a)^2 + 6\phi_a^2\right)\right]. \tag{65}$$

The kinetic coefficient $1/\kappa = 5/4$ follows from the original $\kappa/4$ Lagrangian coefficient only after the momentum convention is fixed. These equations are the explicit algebra underlying the coefficient list received in the supplement [6].

B.3 Variation and boundary jump

Variation of $L_{\mathrm{free}} + \epsilon_R J\phi(R)$, with the canonical counterterm included independently of ϕ, gives Eq. (22). Integrating over a shrinking interval about R removes the regular bulk terms and leaves

$$-\frac{\kappa}{2}R^2[\partial_r\phi]_{R^-}^{R^+} = \epsilon_R J. \tag{66}$$

For the static profile, the jump is $-5x/R^2$. The stationary single-current square gives $x = \epsilon_R J/2$, so the profile jump is $-5\epsilon_R J/(2R^2)$, equal to Eq. (23) at $\kappa = 4/5$. This compatibility is componentwise; it does not infer simultaneous minimization of both noncommuting source-current couplings.

B.4 The spectral square retains the full counterterm

Expanding one polarization's quadratic form in Eq. (26) gives its free spectral energy, both Hermitian interaction terms, and

$$\epsilon_R^2 J_a^2 \int_0^\infty \frac{|u_R(\omega)|^2}{\omega}\mathrm{d}\omega = \epsilon_R^2 J_a^2 \mu. \tag{67}$$

The retained normalization $\mu = 1/(4R)$ therefore returns exactly the counterterm in Eq. (16). Summing both polarizations in a driven lane gives $2g^2\mu$, not $g^2\mu$. No full source energy term is removed by the square completion.

C. Full derivation chain: phase bounds and clock conversion

The supplied exact correlation equation can be written $\dot{z} = -i\Delta z + \eta$, where

$$\eta = 2g(A_\times - iA_+), \qquad |\eta| \le 2|g|\sqrt{A_\times^2 + A_+^2} \le 4|g|\sqrt{\mu E_{f,\mathrm{upper}}}. \tag{68}$$

The last inequality uses the two stated componentwise finite-energy bounds and the complete free-field energy of both polarizations. Since $\nu_{\mathrm{actual}} = \Delta - \mathrm{Im}(\eta/z)$, a positive amplitude lower bound gives Eq. (36). Source–field correlations have been bounded, not factorized away.

For the finite-window certificate, Eq. (38) implies $\Omega t - a(t) \le \gamma(t) \le \Omega t + a(t)$. Division by positive t gives Eq. (39). Subtracting the two envelopes and dividing by $t_2 - t_1 > 0$ gives Eq. (40). None of these operations differentiates an envelope. If the resulting lower mean rate is positive, its reciprocal interval can be used for a finite-window action-per-phase readout. It remains distinct from a certified instantaneous reciprocal.

The kinetic conversion is explicitly

$$\int \mathrm{d}\tau \frac{\kappa}{4} r^2 \dot{\phi}_\tau^2 = \int \frac{\mathrm{d}\gamma}{\nu}\frac{\kappa}{4} r^2 \left(\nu\partial_\gamma\phi\right)^2 = \int \mathrm{d}\gamma \frac{\kappa\nu}{4} r^2 (\partial_\gamma\phi)^2. \tag{69}$$

The gradient and angular terms contain no τ derivative and acquire only $1/\nu$. Source, interaction, and counterterm terms likewise acquire the same $1/\nu$. Finally,

$$\frac{i}{2}\left(\langle\Psi|\dot{\Psi}\rangle - \langle\dot{\Psi}|\Psi\rangle\right)\mathrm{d}\tau = \frac{i}{2}\left(\langle\Psi|\partial_\gamma\Psi\rangle - \langle\partial_\gamma\Psi|\Psi\rangle\right)\mathrm{d}\gamma. \tag{70}$$

This is the retained Berry one-form. The formulas describe the stated coordinate representation along an admissible actual-phase evolution; the state dependence of ν has not been replaced by an independent constant.

Table 5: Coefficient dictionary for the specified canonical action. The same ν enters every phase-coordinate term. The column retaining $\dot{\phi}_\tau$ and the column using $\partial_\gamma\phi$ are different velocity conventions, not competing normalizations.

Term	Model-time coefficient	Phase measure; retains $\dot{\phi}_\tau$	Phase derivative convention
Field kinetic	$\kappa/4$	$\kappa/(4\nu)$	$\kappa\nu/4$
Field radial gradient	$-\kappa/4$	$-\kappa/(4\nu)$	$-\kappa/(4\nu)$
Field angular	$-6\kappa/4$	$-6\kappa/(4\nu)$	$-6\kappa/(4\nu)$
Pair source	$-\epsilon_R$	$-\epsilon_R/\nu$	$-\epsilon_R/\nu$
Boundary interaction	$+\epsilon_R$	$+\epsilon_R/\nu$	$+\epsilon_R/\nu$
Counterterm	$-\epsilon_R^2/(4R)$	$-\epsilon_R^2/(4R\nu)$	$-\epsilon_R^2/(4R\nu)$

D. Full derivation chain: local clock curvature and radial coefficients

For the source clock, $\partial_x E = q_a$ and $\partial_y E = -q_b$. As q_b is independent of x and q_a of y,

$$\partial_x F(E, q_b) - \partial_y F(E, q_a) = q_a F_E(E, q_b) + q_b F_E(E, q_a). \tag{71}$$

Multiplication by the inherited $1/10$ gives Eq. (49); the same coefficient appears in the commutator of the horizontal clock lifts. The adaptive bracket in Eq. (51) belongs to the original path

$$\tan(x/2) = \frac{1}{2} - \frac{s}{6}, \qquad \tan(y/2) = \frac{1}{3} + \frac{s}{6}. \tag{72}$$

Its source report saves midpoint $235/256$ as unexecuted after the deadline. It is not a further evaluated endpoint in this manuscript [4].

For the exterior radial source, the metric readouts give

$$\left(\frac{\mathrm{d}\tau}{\mathrm{d}t}\right)^2 = f - \frac{\beta^2}{f}, \qquad \beta = \frac{\dot{r}}{c}, \qquad f = 1 - A. \tag{73}$$

Composing with $S = -E_0\tau$ gives Eq. (52). The original role coefficient is retained as

$$E_0 = \frac{\hbar\epsilon_{\mathrm{role}}}{t_{\mathrm{SW}}}, \qquad \epsilon_{\mathrm{role}} = \frac{A_0^2 k/2^n}{1 + (A_0/2)(q/2^{D_{\mathrm{space}}})}. \tag{74}$$

Here D_{space} denotes the dimension in that historical expression, not the pair current D. With $m = E_0/c^2$, the source-reported rest Hessians are $L_{AA} = E_0/(4f^{3/2})$ and $L_{vv} = m/f^{3/2} = 4L_{AA}/c^2$ [2].

At the recorded nonzero-speed point, the signed mixed radial response in Eq. (53) enters the complete Schur subtraction. The source report identifies the subtracted mixed payment as $363/125$. The canonical momentum is $\Pi = \pm 1$, the normalized Hamiltonian $h = H/E_0 = 15/16$, and $\beta\Pi - h = -3/5$, retaining the same lapse in both orientations. These are the earlier completed worldline identities; they do not set the later quadrupole-field normalization.

E. Si29 tensors, inverse coordinates, and local limits

The retained Hessians in (u, v) coordinates are given below at their reported decimal precision. They are printed to identify the complete mixed response, not to replace the native rational enclosures [3]:

$$\begin{aligned} G(0.007) &= \begin{pmatrix} 252.754376792 & -785.399243068 \\ -785.399243068 & 17867.308060656 \end{pmatrix}, \\ G(0.013) &= \begin{pmatrix} 110.500345438 & -516.332172340 \\ -516.332172340 & 7083.045979756 \end{pmatrix}, \end{aligned} \tag{75}$$

$$\begin{aligned} H_{\bar{A}}(0.007) &= \begin{pmatrix} 7.753067044 & -6.014821487 \\ -6.014821487 & 445.392955447 \end{pmatrix}, \\ H_{\bar{A}}(0.013) &= \begin{pmatrix} 4.094953599 & -11.510122562 \\ -11.510122562 & -24.841908533 \end{pmatrix}. \end{aligned} \tag{76}$$

The second reference has negative interaction-direction area curvature and positive relative-entropy curvature. The direct proportionality is already excluded at first order; the full Hessians additionally show different directional structure. The reported

reduction of quadratic Taylor residuals by approximately eight when the grid spacing is halved belongs to the saved signed grid, not a fit that removes the linear area response.

The inverse response matrices, with rows (u, v) and columns $(S_{\mathrm{vN}}, \bar{A})$, are

$$B_{\mathrm{inv}}(0.007) = \begin{pmatrix} 0.7205540224 & -20.9812474139 \\ 0.01416981638 & 0 \end{pmatrix},$$
$$B_{\mathrm{inv}}(0.013) = \begin{pmatrix} 0.6384012654 & -6.90877431895 \\ 0.04073674504 & 0 \end{pmatrix}. \tag{77}$$

Their domain is local differential reconstruction. The first-order inverse has finite-step error, explicitly reported in Section 8; no global inverse or numerical neighborhood radius was assigned. The metric $B_{\mathrm{inv}}^{T} G B_{\mathrm{inv}}$ retains mixed entries approximately -3587.67203236 and -342.052506552.

For $w = (1 + 8\nu_{\mathrm{Si}}, \nu_{\mathrm{Si}})$, the recorded contraction is

$$g_{tt}(0.007) = 252.754376792 + 2473.271542528\nu_{\mathrm{Si}} + 21477.200286224\nu_{\mathrm{Si}}^2,$$
$$g_{tt}(0.013) = 110.500345438 + 735.341182326\nu_{\mathrm{Si}} + 5893.753330337\nu_{\mathrm{Si}}^2. \tag{78}$$

The interaction parameter ν_{Si} here is not the actual phase-clock rate ν_{actual} of Section 5. A physical action conversion requires an explicit source/time/boundary map, not a substitution based on a shared symbol.

F. Evidence ledger and source-to-claim register

F.1 Numerical status of the supplied record

Table 6: Evidence categories are kept separate. "Reported complete" refers to the supplied campaign report, not a new reproduction performed while writing this manuscript.

Source	Recorded extent	Status and scope
H02–H06 successor [1]	4,781 original DAG operations; 30 required logarithmic clock calls; 80,940 independent checks	Reported complete, 5/5. Original frozen 0/5 retained. Native predicates/status checks: 481,979. Separate custody pass: 57,291 checks; not added to the mathematical comparison as independent theorems.
Radial action [2]	82 mathematical tasks; 2,485 native operations; 5,418 native checks; 25,447 independent checks	Reported complete. Independent total separates 13,939 mathematical checks and 11,508 provenance checks. Nine retrieval returns are separate.
Si29 [3]	1,116 native operations; two checkpoints; 959 independent checks	Reported complete focused follow-up. Main 833 and follow-up 126 checks. Thirty follow-up operations reuse retained objects without forward-state recomputation.
Clock curvature [4]	17 mathematical tasks; 709 native operations; 1,422 native checks; 4,360 independent checks	Reported complete. Eight retrieval tasks are separate. Original calibration and H02/H03 clock contrast preserved. Zero bracket certifies existence, not uniqueness.
Tensor channels [5]	Ten mathematical jobs; 373 native operations; 746 native checks; 48,226 independent checks	Reported complete. Seven source searches are separate. Observable composition succeeds; the nonzero literal endpoint-action residual is retained.
Interacting action [6]	Joint action; field normalization; charge law; clock-bound and phase-coefficient formulas	Original analytical receiving-law construction; source rules and receiving laws remain distinct.
Write-to-six-clock source [8]	Four source placements; 4×78 incidence; $Q(1)$ spectrum; $E_{\mathrm{ref}} = 573049$; six signed couplings	New source derivation supplied. It describes certified hard-gap and finite-window routes; raw exact lane endpoints and receipt table were not uploaded.
Spin fifth-coordinate completion [9–11]	54 preparations, 1,998 improved upper enclosures; 222 pilot direct/scalar comparisons; 11,214 interval checks	Reported completed J21/F11/T5 source/receiver result, with old lower unchanged; sixth-coordinate extension proposed only.

Source	Recorded extent	Status and scope
Controlled history composition [12]	Six finite histories; 48/48 external symbolic comparisons; 75 wrapper-completed and 14 wrapper-failed tasks	Native history/inverse results supported; failed wrapper-status route explicitly preserved.

F.2 Direct source pointers

The following paths are reproduced from the supplied reports, not presented as newly fetched full files. When a report supplied only relative paths, the missing campaign root is not invented. The older absolute project root is not a portable installation requirement.

H02–H06 numerical successor [1]. The report points to `DERIVATION.md`, `ACTION_LEARNING.md`, `SOURCE_ACTION.md`, `SOURCES_NEW.md`, `INDEPENDENT_DERIVATION.md`, `SEGMENTED_EXECUTION.md`, `RESULT_TABLE.md`, and `SUMMARY.json`. Its verification history retains `verification/H02_1.json`, `verification/H02_2.json`, and the separate V2/V3 path/schema repair records. The report preserves ten source integrals, seven verifier attempts, 41 checkpoint roots, and source-dependency custody.

Radial action and response [2]. Project-relative directory: `GEN4/duck_source_action1_20261014/`. Relevant documents: `DERIVATION_WORK.md`, `SOURCE_INDEX.md`, `WEB_R2_INTERPRETATION.md`, `INDEPENDENT_RESULT.json`, `FINAL_CODE_IDENTITY.json`, `RAW_OUTCOMES.json`, `CODEX_REVIEW.json`, `REPRODUCE.md`, and `FOLLOW_UP.md`. The report identifies the run `GEN4/duck_gen6_local/runs/source-action-voli-20261014-01/` and its completion/resource receipts.

Si29 entropy/area comparison [3]. Relevant relative paths: `DESIGN.md`, `ANALYTIC_NOTES.md`, `GRID.csv`, `native2/RESULT.json`, `INDEPENDENT1.json`, `followup1/RESULT.json`, `FOLLOWUP_INDEPENDENT1.json`, `native2/NATIVE_CHAINS.json`, `native2/SESSION.json`, `native2/CHECKPOINT.json`, `FOLLOWUP.md`, and `RESOURCE_RELEASE.json`. The failed first retrieval remains in `native1/FAILED.json` with its exact code snapshot.

Retained clock ordering and inverse coefficients [4]. Relevant relative paths: `math/RESULT.md`, `math/R2_RESULT_SUMMARY.json`, `sources/COEFFICIENT_MAP.md`, `verification/independent1/VERIFICATION.json`, `math/SUCCESSFUL_CHAINS.json`, `EXECUTION_CUSTODY.json`, `math/PREPARATION_HISTORY.json`, `ops/PLAN_BUILDER_CUSTODY_REPAIR.json`, `ops/benchmark_results/RESULT.json`, `ops/ACTUAL_COSTS.json`, and `sources/SOURCE_CHANNEL_FOLLOWUP.md`.

Write phases, tensor channels, and lifted inverse [5]. Relevant relative paths: `math/RESULT.md`, `math/R2_RESULT_SUMMARY.json`, `math/SUCCESSFUL_CHAINS.json`, `verification/independent1/VERIFICATION.json`, `EXECUTION_CUSTODY.json`, `RAW_OUTCOMES.json`, `sources/ACTION_UNIT_MAP.md`, `ops/RESOURCE_NOTE.md`, `ops/ACTUAL_COSTS.json`, and `FOLLOW_UP.md`. These relative names belong to this campaign and must not be merged with identically named files from [4].

Complete interacting action [6]. The author-supplied text identifies H002624 as its upstream source-sector and six-lane result. It provides the equations and derivation headings reproduced here, but no new cycle result table, raw lane spectra, or exact resolved-clock gzip endpoints. Those inputs must be received from their proper source before numerical receiving certificates are added. H002624 is not equated with another report number by inference.

Original Write spectrum and six clock receivers [8]. The supplied `DERIVATION(4).md` locates the source and receiving programs under `SAM_MATTER_LEARNING/generated_bridge.py`, `SAM_MATTER_LEARNING/isotope_interaction.py`, and their native counterparts. The actual receiving radius is said to be pinned at `samland_joint_response1/INPUT.json`. The source names `clock/DERIVATION.md` and separate corrected and original native graphs; their actual absolute root, interval tables and certified root files have not been fabricated here.

Spin fifth-coordinate source and receiver [9–11]. The scalar map and original-source receiver are in the supplied `DERIVATION(2).md` and `DERIVATION(3).md`. The project report locates the new full-source evaluations under `GEN4/spin_upper_source1_20261014/pilot3/V000` and `GEN4/spin_upper_source1_20261014/native1/`, with `SOURCE_RESULTS.json` in the source parent. Its Duck run is `GEN4/duck_gen6_local/runs/spin-upper-20261014-01`, and its response/receipt files include `RESPONSE_BOUNDS.csv`, `SOURCE_ACTION_SPLIT.json`, `RAW_OUTCOMES.json` and `FOLLOW_UP.md`. These are source-reported links, not locally audited full original files.

F.3 What must accompany a complete executable release

The supplied reports provide a strong source map but not a standalone computational distribution. A reproducible release would attach the pinned source matrices and phase words, exact interval endpoints, hard/soft spectral inputs, full lane assignments, numerical manifests, run/checkpoint receipts, and executable independent checkers cited above. This manuscript supplies neither guessed replacements nor an assertion that those underlying files were audited here. Its source package is an editorial reconstruction of the supplied record.

The new source derivation specifies the functional u_R, $b = 1/(6\kappa\pi)$, $b_3 = R/42$, and the adopted split $\lambda = 64\Delta$; it also identifies the conditional gap/amplitude and phase-envelope construction. The exact original source-current certificate roots, all finite-window endpoints, and the complete numerical table still have to accompany a full executable release. They are not generated from rounded q_a values. Likewise, the spin report's exact rational endpoints reside in its named response table, not in its printed range summary.

G. Notation and coefficient provenance

Table 7: Context-sensitive notation. Descriptive aliases prevent unrelated source quantities from being silently identified.

Symbol	Meaning in this manuscript	Source/provenance distinction
ϕ_a, ϕ_b, ϕ_p	Lifted original endpoint/partner phases	Not the radial field functions $\phi_+, \phi_\times$.
γ, δ, χ	Receiving phase coordinates in Eq. (1)	χ is the retained common phase here; the radial lapse is named explicitly instead.
$\gamma = -\arg z$	Interacting lane phase, continued from its retained initial value	Actual trajectory phase; its rate need not equal Δ or Ω.
ω_{Write}	Original legal phase-action one-form	Distinct from the full quantum variational action and from positive clock duration.
κ_{clk}	Local coefficient of $\mathrm{d}\alpha_{\mathrm{clock}}$	The source clock report calls this κ; it is not the quadrupole field constant.
$\kappa = 4/5$	Field normalization from the static profiles and $B = 2I_2$	Conditional on the stated TT metric, quadrupole law, and full interior/exterior matching.
$B = 2I_2$	Literal TT tensor metric	Not the Si29 differential inverse $B_{\mathrm{inv}} = J^{-1}$.
R	Retained boundary radius in the source–field action	Not SAM route radix; $R_H = 1$ is separately reported for the earlier clock calibration.
D, C	Noncommuting pair currents	Not the spatial dimension D_{space} or relative entropy $\mathcal{D}$.
K_{pair}	$E_{\mathrm{even}} - I$, commuting pair-energy remainder	No identification is made with the preparation account K in the latest certificate.
ϵ_R	Retained source/boundary receiving scale	Not the earlier constraint multiplier or G471 role factor. No numerical value is assigned here.
ϵ_{role}	G471 role factor entering E_0	The earlier conditional clock one-form uses a separate declared multiplier, with probe value one.
I_s	Action unit of the joint variational law	The phase-convention identification $I_s = \hbar$ is not a numerical determination of units.
$\Delta = 2\epsilon_R$	Bare lane gap	Distinct from the stationary hard-field gap Ω.
ν_{actual}	Actual correlated phase rate	All phase-action coefficients share this same dependent quantity.
ν_{Si}	Si29 interaction parameter in $Q \otimes I + I \otimes G + \nu_{\mathrm{Si}} Q \otimes G$	Not a clock rate.
A, $\bar{A}$, $\mathcal{A}$	Accumulation; normalized enclosing area; preparation-bound parameter	Three different roles, separated explicitly in Sections 5, 7, and 8.
E, $\mathcal{E}(t)$	Earlier source contact energy; soft remainder bound	Neither is silently substituted for total interacting energy.
Q_{ch}	Signed angular charge $N_s + N_+ - N_-$	Distinct from the single-source matrix Q in Q_L, Q_R.
$\mathsf{B}, T, K, Q(g_H)$	Write incidence, source Gram, signed response and receiving matrix	The K response is inherited from a completed source map; $g_H = 1$ is a specified diagnostic receiving chart.
$E_{\mathrm{ref}}, \lambda_{\mathrm{ref}}$	Full reference preparation energy and role-capacity scalar	Reference energy 573049 is exact in the $g_H = 1$ chart; dimensional conversion retains t_{SW}.

Symbol	Meaning in this manuscript	Source/provenance distinction
$\epsilon_{\text{current}}$	Signed active-Write current rate scale	Not the autonomous ϵ_R of the joint Hamiltonian.
$A_{\text{spin}} = N(I - P)$	Frustrated-spin source operator	Not the SAM accumulation field $A(r)$ or the finite source matrix $Q(g_H)$.
J_{21}, F_{11}	Original spin source contractions	Not fitted receiver coefficients; needed in the spin 5×5 response.

H. Original Write four-source spectrum, reference energy and clock receiver

The four by seventy-eight signed incidence B is admitted by the original single-address bit-flip boundary. Its Gram $T = \mathsf{B}\mathsf{B}^*$ and the completed signed source response K are the exact matrices of Eqs. (8) and (9). In the row order of the four labelled cube paths, the unit-strength receiving matrix is

$$Q(1) = T + K = \begin{pmatrix} 756 & 104 & 40 & 0 \\ 104 & 2100 & 0 & 0 \\ 40 & 0 & 756 & 40 \\ 0 & 0 & 40 & 756 \end{pmatrix}. \tag{79}$$

Writing $x = q - (84 + 672 g_H)$ yields (11). A symbolic characteristic-determinant calculation using the displayed matrices confirms this identity identically as a polynomial in q, g_H. For the specified receiving chart $g_H = 1$, its real roots are the four decimals of Table 1. The completed native campaign retains their exact rational certificates; this manuscript prints only displays, not fabricated certified endpoints.

The full sixteen-dimensional source is the tensor square of the four-dimensional Q representation with a retained Swap term. For each of the six unordered pairs $a < b$, the two-state source lanes have diagonal even part $q_a + q_b + q_a q_b$, exchange splitting ± 1, and signed current magnitude $d_{ab} = q_a - q_b$. For the four diagonal states $|aa\rangle$, the source energy is $1 + 2q_a + q_a^2$. The retained reference $|00\rangle$ has $E_{\text{ref}} = 573049$; it is not permissible to discard its even term when matching to a clock.

The original capacity law is

$$\lambda_{\text{ref}} = \frac{A_0^2 k / 2^n}{1 + (A_0/2)(q_{\text{role}}/2^{D_{\text{space}}})}, \qquad E_{\text{ref}}^{\text{phys}} = \frac{\hbar \lambda_{\text{ref}}}{t_{\text{SW}}}. \tag{80}$$

The representation $I_s \epsilon_R E_{\text{ref}} / t_u = E_{\text{ref}}^{\text{phys}}$ gives Eq. (13). With $I_s = \hbar$, $t_u = t_{\text{SW}}/\lambda_{\text{ref}}$, its right side equals 573049^{-1}. The retained signed active-Write current in Eq. (14) is a separate dependent quantity. Matching the scalar Write action and an occupied quantum source action would additionally require

$$I_s \langle G \rangle \dot{\gamma} = \hbar(\dot{\gamma} + 3\dot{\chi}). \tag{81}$$

This is a source compatibility question, not a silent identification of the two one-forms. The equality of R_H and the quadrupole R, the physical choice of g_H and the original boundary incidence remain separately assigned. The exact initial correlation of Eq. (46) supplies a phase receiving check within those assignments, with all four counter-rotating words, both helicities and the even source energy still present.

I. Fifth-coordinate spin Schur receiver and the source-action excess

The finite frustrated-spin source uses a self-adjoint $A_{\text{spin}} = N(I - P)$ with original Gibbs metric and fixed attempt clock. Let $b_4 = (d, e, u_d, u_e)$ and $b_5 = (b_4, w_d)$. Retain earlier $G, D, W, G_5, D_5, \xi, \gamma$ and $H = G^{-1}D$. The additional source operands are

$$J_{21} = \langle w_e, A_{\text{spin}} w_d \rangle, \qquad F_{11} = \langle A_{\text{spin}} w_d, A_{\text{spin}} w_d \rangle. \tag{82}$$

For $j = J_{21}, f = F_{11}$, define $t = H^T k + E(J_{11}, j)^T$, $T_4 = DG^{-1}D + EWE^T$, $K_5 = T_5 + (\xi - \gamma)D_5 - \xi\gamma G_5$ with

$$T_5 = \begin{pmatrix} T_4 & t \\ t^T & f \end{pmatrix}, \quad r_5 = (D_5 - \gamma G_5) e_d. \tag{83}$$

The full inverse receiver calculates

$$U_5 = \frac{\mathrm{Var}(d) - r_5^T K_5^{-1} r_5}{\xi + \gamma}. \tag{84}$$

The independently rearranged scalar route writes $B(j) = B_0 + jB_1$, $C(f) = f + (\xi - \gamma)J_{11} - \xi\gamma W_{11}$ and the Schur complement

$$\begin{aligned} S(j, f) &= C(f) - B(j)^T K_4^{-1} B(j) \\ &= f + s_0 + s_1 j + s_2 j^2, \\ U_5 &= U_4 - \frac{(a_0 + a_1 j)^2}{(\xi + \gamma) S(j, f)}. \end{aligned} \tag{85}$$

Here $a_0 = B_0^T K_4^{-1} r_4$, $a_1 = B_1^T K_4^{-1} r_4$ and the s_i are the exact source-independent coefficients received in the supplied scalar derivation [9, 10]. The reported positivity of S and the gain for all $1,998$ histories makes the bound reduction an actual received result, not an algebraic promise.

The Gram/projection admissibility condition, distinct from the history-dependent upper target, is

$$f \geq h + \frac{(j - \alpha J_{11})^2}{\delta}, \qquad \alpha = \frac{W_{21}}{W_{11}}, \quad \delta = W_{22} - \frac{W_{21}^2}{W_{11}}, \quad h = k^T G^{-1} k + \frac{J_{11}^2}{W_{11}}. \tag{86}$$

The report's source-action split introduces

$$F_{\text{excess}} = F_{11} - \left[h + \frac{(J_{21} - \alpha J_{11})^2}{\delta} \right]. \tag{87}$$

In six source pilots it is positive and approximately 1.4209363–1.4212459. A direction-factor interval close to unity and an independent source-action factor near 0.5477 identify the unfinished *source-action residual*, rather than another rotation of the already aligned payment direction. The suggested next vector $r_{\text{next}} = A_{\text{spin}} w_d - P_{b_5}(A_{\text{spin}} w_d)$ belongs to the future sixth-coordinate extension. It would require an additional action contraction and squared-action norm; neither is claimed as executed here.

For source case V000_BASE, the printed approximate moments are $J_{21} \in [1.0532051 \times 10^{-5}, 7.4632747 \times 10^{-5}]$ and $F_{11} \in [6.3832583, 6.3834661]$. They are representative displays only, not replacements for the certified original enclosures in `SOURCE_RESULTS.json`. As in the original report, all paid background, interpolation and semigroup terms remain in the source calculation and the lower L_5 is preserved unchanged.

Version close. The completed source campaigns and their failures/successors remain distinct. This version additionally receives the admitted original-Write source spectrum and the H002626-to-original-source J21/F11/T5 completion. The six-lane hard/soft receiving formulas and spin upper results retain their own source assignments; a universal physical bulk/boundary coupling or an external matched performance benchmark is not inferred.